

Joint Minorants for Nine Points and Simple Critical Zeros of the Riemann Zeta Function

RH–Weil research project

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Abstract

We improve a fixed eight-point certificate for a perturbed Montgomery–Taylor window by jointly treating two overlapping frames. The two frames share their actual distance-square variables and all common gap and span constraints. From the existing continuous certificate we capture 70 low-slack cells, restore their reflections, and reduce the joint problem to 289 labelled intersections. Rational epigraph duals, checked with exact finite-box residual corrections, pay the local reward $805103/10^8$ on every intersection. Together with the existing lossless separation and all-zero spectral assembly this gives

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194897} = 0.6735476624367078 \dots$$

The denominator counts all nontrivial zeros with multiplicity; the numerator counts simple zeros on the critical line. Off-line zeros are retained. The analytic argument assumes no zero-free strip or RH. The finite computational input retains the disclosed PC8CL continuous-audit and compiled-integer trust scope. The new dual verifier uses only integer and rational arithmetic; numerical linear programming merely locates witnesses. This is not a complete Lean kernel proof.

1 Statement, provenance and comparison

Let $N(T)$ count the nontrivial zeros ρ of ζ , with $0 < \Im \rho \leq T$, including multiplicity. Let $N_0^s(T)$ count the simple zeros among them with $\Re \rho = 1/2$.

Theorem 1.1. *Using the continuous PC8CL certificate stated in Input 2.1 and the exact joint certificate established below, one has, unconditionally,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194897}. \quad (1)$$

No zero-free strip, Riemann hypothesis or higher correlation conjecture is assumed.

The window and eight-point data are prior inputs from the fixed public AM construction [4]. Their lossless assembly [6] gave $66812491/99194997$. The present improvement is in the local certificate: two overlapping frames share the actual kernel squares and all their continuous linear span constraints. We adapt the separation mechanism used by Knausgård [5, Section 5], without consuming that paper’s headline native computation axiom. The new rational bound exceeds the previous repository bound by

$$\frac{6681249100}{99194897 \cdot 99194997} > 0,$$

approximately 0.0000679014 percentage points. Its comparison with the fixed Knausgård v1 benchmark is

$$\frac{66812491}{99194897} - \frac{1669159}{2478195} = \frac{2326052122}{245824297770915} > 0. \quad (2)$$

This is a fixed-source comparison, not an exhaustive global priority claim. A zero-counting proportion is not a measure of progress toward a proof of RH, and the separate zero-free-region bounds are unchanged.

2 Window and admitted local certificate

Put $\beta = 1/\sqrt{2}$ and $Z_0 = \sqrt{2} \sin \beta$. On $I = [-1/2, 1/2]$ define

$$f(u) = \frac{\cos(\sqrt{2}u) + \sum_{j=1}^{12} c_j \cos(2\pi j u)}{Z_0}, \quad K(t) = \int_I f(u) e^{-2\pi i t u} du. \quad (3)$$

Extend f by zero outside I . The integer numerators $10^9 c_j$ are

$$(12310798, -15041681, 3867664, 6489926, -992327, 5580097, \\ -6846472, 3781297, -5670353, 3355089, -450523, -218483).$$

Thus $S = \sum_j |c_j| = 6460471/10^8 < 13/200$. Since $\cos(\sqrt{2}u) \geq 1 - u^2 \geq 3/4$ on I , the numerator is positive. The integer cosine modes integrate to zero, so f is an even probability density. Also

$$Z_0 = \int_I \cos(\sqrt{2}u) du \geq 11/12. \quad (4)$$

For eight ordered points $y_0, \dots, y_7$, write $g_r = y_{r+1} - y_r \geq 0$, $0 \leq r \leq 6$. Set

$$c_0 = \frac{805003}{10^8}, \quad c = \frac{805103}{10^8}, \quad B = \sum_r b_r = \frac{404350}{10^8},$$

$$(10^8 b_r)_{r=0}^6 = (28898, 57272, 75526, 80958, 75526, 57272, 28898).$$

All pair weights a_{ij} are nonnegative. Table 1 gives their integer numerators.

$i \setminus j$	1	2	3	4	5	6	7
0	13630830	0	19565904	20317441	45030671	100000000	200000000
1		29997996	30621878	47766489	79682558	109938656	100000000
2			37356140	69378121	65335210	79682558	45030671
3				38030065	69378121	47766489	20317441
4					37356140	30621878	19565904
5						29997996	0
6							13630830

Table 1: The weights $10^8 a_{ij}$ of the fixed AM certificate.

Computational input 2.1 (Fixed PC8CL certificate). *For every $(g_0, \dots, g_6) \in [0, \infty)^7$,*

$$\sum_{r=0}^6 b_r g_r + \sum_{0 \leq i < j \leq 7} a_{ij} K \left(\sum_{r=i}^{j-1} g_r \right)^2 \geq c_0. \quad (5)$$

This is a continuous real-gap assertion, not a grid sampling claim. Its provenance, finite replay and continuous soundness evidence are specified in Section 7. We use the complete admitted certificate, rather than the upstream final zeta theorem. Direct integer summation of Table 1 gives

$$10^8 \left(\sum_{i=0}^{7-s} a_{i, i+s} \right)_{s=1}^7 = (199999997, 199999998, 199999996, 199999998, \\ 199999998, 200000000, 200000000). \quad (6)$$

In particular each span-length budget is at most two.

3 The joint continuous nine-point certificate

Write $F_8(h)$ for the left side of (5) and put $\theta = 4/5$, $\delta = 10^{-6}$, so $c = c_0 + \delta$. For $g = (g_0, \dots, g_7) \in [\theta, \infty)^8$, let $h^- = (g_0, \dots, g_6)$ and $h^+ = (g_1, \dots, g_7)$. Define

$$F_9(g) = \frac{1}{2}\{F_8(h^-) + F_8(h^+)\} + 2K(g_0 + \dots + g_7)^2. \quad (7)$$

The new pair weights are

$$a'_{ij} = \frac{1}{2}a_{ij}\mathbf{1}_{j \leq 7} + \frac{1}{2}a_{i-1, j-1}\mathbf{1}_{i \geq 1} + 2\mathbf{1}_{(i,j)=(0,8)},$$

where out-of-range old weights are zero. The gap fees are

$$(10^8 b'_r)_{r=0}^7 = (14449, 43085, 66399, 78242, 78242, 66399, 43085, 14449).$$

All weights and fees are nonnegative; $\sum b'_r = B$. Each index-span budget is at most two, including span eight.

3.1 A complete low-slack outer cover

Put $\mathcal{L} = \{h \in [\theta, \infty)^7 : F_8(h) < c_0 + 2\delta\}$. If either frame is outside $\mathcal{L}$, Input 2.1 and (7) already give $F_9 \geq c$. We explain why the captured cells cover all of $\mathcal{L}$. The old small-gap domain $h_i \leq 1/2$ is absent. In any old clear COV interval, a single adjacent-pair term already pays c_0 ; the pressure adds at least $\theta B = 8087/2500000 > 2\delta$. In any old large-gap domain, one fee pays c_0 and the remaining fees add at least $\theta(B - \max b_i) = 5053/1953125 > 2\delta$. Thus only the bad COV intervals and their 32 root trees remain.

In the half-space $h_6 \geq h_0$, the old direction leaves $u_6 < l_0$ cannot contain the point. Farkas empty-cell leaves do not contain it either. Every remaining successful leaf supplies an interval minorant. Keep the original reward, return value and packed bit cursor; copy only its final reward comparison with the new target $805203/10^8 = c_0 + 2\delta$. A passing stronger leaf excludes the point from $\mathcal{L}$; a nonpassing stronger leaf with a passing old comparison is captured. The actual unchanged 32 root consumption positions and seven outer COV checks all pass. This captures exactly 70 cells. Their full reflections, using $h_r \mapsto h_{6-r}$ and $(i, j) \mapsto (7-j, 7-i)$ on spans, restore the other half-space. There are 140 source labels and 132 distinct geometric cells. These are outer cells, not subsets on which every point necessarily has low slack.

Each cell retains seven gap and 21 long-span closed bounds, in units $SC = 32768$. A left/right intersection retains all eight gaps and 27 distinct long spans. Prefix coordinates turn these into exact closed difference constraints. In units $5SC$ the lower gap bound θ is exactly the integer 131072. Negative cycles exclude empty intersections; shortest-path potentials verify feasibility otherwise. The 19600 labelled pairs reduce to 422 after common gap bounds and to 289 after all common spans and the complete closure. Labels are retained even when their geometric cells coincide, since their minorants can differ. There is no integer rounding of a real gap.

The capture and continuous semantics reuse the admitted fixed PC8 input. The unchanged old reward comparisons, root cursors and atom validity checks are essential. A failed stronger comparison alone would not prove a counterexample, and the list of cells alone would not prove its completeness.

3.2 Valid individual tangent cuts

Let $w(t) = K(t)^2$. For an enabled point $p \in [l, u]$, the old continuous tangent assertion gives real v, d with

$$w(x) \geq v + d(x - p) \quad (l \leq x \leq u), \quad v \geq v_0, \quad d_- \leq d \leq d_+.$$

These computations are finite computational evidence in the explicitly stated integer and rational trust scope; they are not consequences of floating-point solver status. The span-eight variable has no positive cut in this certificate, so the joint old 33 pair terms already suffice.

Proposition 3.3 (Uniform local improvement). *Under the admitted continuous PC8 atom and capture semantics, $F_9(g) \geq c = 805103/10^8$ for every $g \in [4/5, \infty)^8$.*

Proof. Outside the two-frame low-slack case, the first observation in this section pays c . Inside it, completeness of the reflected outer cover places the actual point in at least one of the 289 labelled intersections. Its actual gaps and squares satisfy the associated matrix and box. The exact dual bound pays c there. This covers the entire unbounded separated domain, including every closed boundary. $\square$

Lemma 3.4 (Direct sliding). *For any m real points separated by at least θ ,*

$$\sum_{i \neq j} K(y_i - y_j)^2 \geq c(m - 8) - B \text{span}(y). \quad (8)$$

The span is zero when $m \leq 1$.

Proof. For $m \geq 9$ sum Proposition 3.3 over the $m - 8$ nine-point windows. Each pair of a fixed index span receives total nonnegative weight at most two. Each adjacent gap receives total fee at most B . The pair sum is at most the full ordered pair sum, and the total pressure is at most $B \text{span}(y)$. For $m \leq 8$ the right side is nonpositive. $\square$

4 A band-limited majorant for the same window

We use $\text{sinc } t = \sin(\pi t)/(\pi t)$, with $\text{sinc } 0 = 1$, and put

$$\theta = \frac{4}{5}, \quad \gamma = \frac{61}{100}, \quad g(u) = \gamma \{\text{sinc}(\theta(u - 1/2)) + \text{sinc}(\theta(u + 1/2))\}^2. \quad (9)$$

Lemma 4.1. *The nonnegative function g satisfies $g \geq f$ on $\mathbb{R}$, $\widehat{g}(t) = 0$ for $|t| \geq \theta$, and*

$$\Lambda := \int_{\mathbb{R}} g(u) du = \frac{2\gamma}{\theta} (1 + \text{sinc } \theta) \leq \frac{61}{32} < 2. \quad (10)$$

Proof. Each shifted sinc has a rectangular Fourier transform supported in $[-\theta/2, \theta/2]$. The square in (9) therefore has Fourier support $[-\theta, \theta]$. The convolution is continuous and zero at both boundary points, so the conclusion includes equality $|t| = \theta$. Parseval gives the displayed formula for Λ . As $\sin(4\pi/5) = \sin(\pi/5) \leq \pi/5$, $\text{sinc}(4/5) \leq 1/4$, proving its stated upper bound.

The remaining assertion $g \geq f$ is proved by the small exact rational certificate in Appendix A. Specifically it establishes

$$g(k/256) - f(k/256) > 9/100 \quad (0 \leq k \leq 128), \quad \sup_{u \in I} |(g - f)'(u)| < 16.$$

Both functions are even, and every point of $[0, 1/2]$ is at distance at most $1/512$ from one of these nodes. Hence on I

$$g(u) - f(u) > 9/100 - 16/512 = 47/800 > 0.$$

Outside I the assertion follows from $f = 0$ and $g \geq 0$. The derivative estimate is the essential continuous bridge; the grid inequalities alone would not suffice. $\square$

Lemma 4.2 (Close-pair reward). *For $|t| \leq 4/5$, one has $K(t) > 134/825$ and $K(t)^2 > 1/40 > c$.*

Proof. For $0 \leq t \leq 4/5$, the functions $\cos(\sqrt{2}u)$ and $\cos(2\pi tu)$ both decrease on $[0, 1/2]$. Chebyshev's integral covariance inequality, divided by Z_0 , gives for the unperturbed kernel $K_0(t) \geq \text{sinc } t \geq \text{sinc}(4/5)$. The last monotonicity follows by differentiating $\sin x/x$ on $[0, \pi]$. Using $\pi^2 < 10$ and $\sin x \geq x - x^3/6$ at $x = \pi/5$ gives $\text{sinc}(4/5) > 7/30$. The perturbation of the normalized kernel has modulus at most $S/Z_0 < 39/550$. Thus $K(t) > 7/30 - 39/550 = 134/825$. Finally $(134/825)^2 = 17956/680625 > 1/40 > c$. Evenness covers negative t . $\square$

5 Lossless spectral assembly

Choose even $\chi_\varepsilon \in C_c^\infty((-1/2, 1/2))$, $0 \leq \chi_\varepsilon \leq 1$, tending to one on the interior, and set

$$m_\varepsilon = \int_I f \chi_\varepsilon^2, \quad f_\varepsilon = f \chi_\varepsilon^2 / m_\varepsilon, \quad K_\varepsilon = \widehat{f_\varepsilon}, \quad d_\varepsilon = \|f_\varepsilon - f\|_1.$$

Then $m_\varepsilon \rightarrow 1$, $d_\varepsilon \rightarrow 0$, and $f_\varepsilon \rightarrow f$ in $L^1 \cap L^2$. The square root $\eta_\varepsilon = \sqrt{f_\varepsilon}$ is smooth, even, compactly supported in the interior of I . On the real axis, $|K_\varepsilon - K| \leq d_\varepsilon$ and both kernels have modulus at most one. Throughout the argument fix ε sufficiently small that $m_\varepsilon > 61/64$.

For a finite set Y of n real points let $U = (K_\varepsilon(y_i - y_j))_{i,j}$ be its positive Gram matrix, with unit diagonal. Define

$$\phi_2(t) = t^2 - (t - 2)_+^2, \quad J(U) = \text{tr } \phi_2(U) - n.$$

The scalar function ϕ_2 is convex on $[0, \infty)$.

Proposition 5.1. *For every finite Y ,*

$$J(U) \geq cn - B \text{span}(Y) - 8c - 32d_\varepsilon n. \quad (11)$$

Proof. Select a maximal collection of disjoint pairs with distance strictly less than θ . The remaining set S is θ -separated. For any coefficient vector (z_j) on S , the pointwise majorant gives

$$\int f_\varepsilon(u) \left| \sum_{j \in S} z_j e^{-2\pi i y_j u} \right|^2 du \leq \frac{1}{m_\varepsilon} \int g(u) \left| \sum_{j \in S} z_j e^{-2\pi i y_j u} \right|^2 du = \frac{\Lambda}{m_\varepsilon} \sum_{j \in S} |z_j|^2.$$

The cross terms vanish even at separation exactly θ . Thus $0 \preceq U_{SS} \preceq (\Lambda/m_\varepsilon)I \prec 2I$. Normalization by m_ε is necessary here. Consequently $\phi_2(U_{SS}) = U_{SS}^2$.

Write $m = |S|$. Apply each local certificate to K and transfer only its finitely many weighted pair terms to K_ε . Each square changes by at most $2d_\varepsilon$; the total weight in one nine-point window is at most 16 by the span budgets. The sliding argument, including the cases $m \leq 8$, yields

$$\text{tr } \phi_2(U_{SS}) - m \geq cm - B \text{span}(S) - 8c - 32d_\varepsilon m.$$

Each removed pair has a Gram matrix with eigenvalues $1 \pm |K_\varepsilon(t)|$ in $[0, 2]$. Its contribution to J is $2K_\varepsilon(t)^2 \geq 2c - 4d_\varepsilon$, by Lemma 4.2.

Diagonalize the principal block for S and each two-point block. Scalar Jensen in the resulting orthonormal basis proves $\text{tr } \phi_2(U) \geq \sum_{\text{blocks}} \text{tr } \phi_2(U_{\text{block}})$. This requires scalar convexity, not operator convexity. Summing the pair rewards and the separated-set reward gives (11), since $\text{span}(S) \leq \text{span}(Y)$ and the pair error is at most $2d_\varepsilon(n - m) \leq 32d_\varepsilon(n - m)$. Empty or small separated sets are covered by the same single boundary charge $8c$. $\square$

6 All-zero Hermitian counting and the energy constant

6.1 The unconditional analytic input

For each fixed ε , the unconditional correlation formula of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [1, Lemma 5], with the correction described in [2, Section 3], gives

$$\sum_{0 < \Im \rho, \Im \rho' \leq T} K_\varepsilon \left(i(\rho - \rho') \frac{\log T}{2\pi} \right)^2 = (C(f_\varepsilon) + o_\varepsilon(1))N(T), \quad (12)$$

where

$$C(h) = \int_I h(u)^2 du + \int_I \int_I |u - v| h(u) h(v) du dv. \quad (13)$$

The sum includes multiplicities and all off-line zeros. For completeness, the weighted lemma applied to a fixed real even test q supported in $[-1, 1]$ has weight $w(\rho - \rho') = 4/(4 - (\rho - \rho')^2)$ and main term $q(0) + 2 \int_0^1 uq(u) du$. Apply it separately to the two fixed smooth functions $q = f_\varepsilon * f_\varepsilon$ and q'' . For $z = i(\rho - \rho') \log T/(2\pi)$, integration by parts gives

$$\left(\widehat{q}(z) - \frac{\widehat{q}''(z)}{4(\log T)^2} \right) w(\rho - \rho') = K_\varepsilon(z)^2.$$

The second fixed-test contribution is $O_\varepsilon(N/(\log T)^2)$; the first gives (12). The lemma allows signed tests, so its use for q'' is legitimate. This is the fixed-smoothing argument also used in [3]. No estimate uniform in a T -dependent smoothing parameter is used.

6.2 Retaining off-line zeros

Put $L = \log T$ and $z_\rho = -i(\rho - 1/2)L/(2\pi)$. The multiset of these points is conjugation-invariant. First in the complex Hilbert space $L^2(I; \mathbb{C})$ set $v_z(u) = \eta_\varepsilon(u)e^{-2\pi i z u}$. For each distinct nonreal conjugate pair write $g_z = (v_z + v_{\bar{z}})/2$ and $h_z = (v_z - v_{\bar{z}})/(2i)$. The vectors v_x for real x , together with g_z and h_z , belong to the real Hilbert space $\mathcal{H}_\mathbb{R} = \{v : \overline{v(u)} = v(-u)\}$: the conjugate reflection J satisfies $Jv_z = v_{\bar{z}}$. Define the finite self-adjoint operator on their real span by

$$A_T = \sum_{\text{distinct real } x} \mu_x v_x \otimes v_x + 2 \sum_{\text{distinct nonreal pairs}} \mu_z (g_z \otimes g_z - h_z \otimes h_z).$$

It acts on the finite span of these real-type vectors. The identity $\|g_z\|^2 - \|h_z\|^2 = 1$ and direct tensor expansion give

$$\text{tr } A_T = N(T), \quad \text{tr } A_T^2 = (C(f_\varepsilon) + o_\varepsilon(1))N(T). \quad (14)$$

Indeed the Hilbert–Schmidt square expands to $\sum_{z,w} K_\varepsilon(z - \bar{w})^2$; conjugation reindexes this as the sum in (12). Individual complex terms need not be positive.

Let P be the sum of the unit features at simple real points, $n = N_0^s(T)$, and $Q = A_T - P$. If r is the number of repeated real points and k the number of nonreal conjugate pairs, then

$$n_+(Q) \leq b := r + k, \quad N(T) \geq n + 2b.$$

Each remaining real point has multiplicity at least two; each nonreal pair contributes at least two zeros and at most one positive rank direction. No location-based deletion has occurred.

Lemma 6.1 (Finite spectral counting). *If $A = P + Q$, P is the sum of n unit features with Gram U , $n_+(Q) \leq b$, $\text{tr } A = N$ and $N \geq n + 2b$, then*

$$\text{tr } A^2 \geq 2(N - n) + \text{tr } \phi_2(U) = 2N - n + J(U). \quad (15)$$

Proof. Enlarge the finite ambient space by zero directions if necessary. If $\lambda_i(P) = p_i$ are its n Gram eigenvalues, including zeros, minmax gives $\lambda_{b+i}(A) \leq p_i$. For $t \leq p$, $p \geq 0$, $4t - t^2 \leq 4p - \phi_2(p)$; also $4t - t^2 \leq 4$ for all real t . After the first b eigenvalues use the first inequality; any remaining eigenvalues are nonpositive and contribute at most zero. Since $\sum p_i = n$, this proves $4N - \text{tr } A^2 \leq 4b + 4n - \text{tr } \phi_2(U)$. Now $4b \leq 2(N - n)$ gives (15). Zeros in the n -dimensional Gram spectrum are included once; the added ambient zero directions contribute nothing. $\square$

6.3 Energy and completion of the proof

Define $(\mathcal{T}h)(u) = h(u) + \int_I |u-v|h(v) dv$. For $f_0 = \cos(\sqrt{2}u)/Z_0$, $(\mathcal{T}f_0)'' = 0$ and evenness makes $\mathcal{T}f_0$ constant. Therefore a zero-mass perturbation has zero cross energy with f_0 . Orthogonality of the integer cosines, and $\mathcal{T} \cos(2\pi ju) = (1 - 1/(2\pi^2 j^2)) \cos(2\pi ju) + \text{constant}$, give

$$C(f) = \frac{1}{2} + \beta \cot \beta + \frac{\sum_{j=1}^{12} c_j^2 (1/2 - 1/(4\pi^2 j^2))}{2 \sin^2 \beta}. \quad (16)$$

The exact rational computation in Appendix A establishes

$$2 - C(f) > \frac{67216841}{10^8}. \quad (17)$$

In particular the energy cost of the perturbation is paid; the unperturbed energy constant is not substituted.

Proof of Theorem 1.1. The simple real positions lie in $(0, X_T]$, with $X_T = T \log T / (2\pi)$ and $X_T/N(T) \rightarrow 1$ by Riemann–von Mangoldt. Apply Proposition 5.1 to their Gram and then Lemma 6.1:

$$(1 - c + 32d_\varepsilon)n \geq 2N(T) - \text{tr } A_T^2 - BX_T - 8c.$$

Fix sufficiently small ε , let $T \rightarrow \infty$ using (14), and only then let $\varepsilon \rightarrow 0$. The $L^1 \cap L^2$ convergence implies $C(f_\varepsilon) \rightarrow C(f)$ in (13). Consequently

$$\liminf_T \frac{n}{N(T)} \geq \frac{2 - C(f) - B}{1 - c} > \frac{67216841 - 404350}{10^8 - 805103} = \frac{66812491}{99194897}.$$

The weak inequality stated in the theorem is therefore valid. The boundary charge $8c$ disappears only after division by $N(T)$. $\square$

7 Verification scope and further improvement

Input 2.1 is pinned to the public `dani-bound-2026` source [4] at commit

`d272437e7eb17b87c8c3d7cece592aa23785ab`.

Its raw `Solution.lean` is 1,675,641 bytes with SHA-256

`012c6ac5f9282158a686500dc0c967bf0a9b00e1a6192734d8f237bb23dd2d5f`.

The repository admission combines a continuous interval-soundness audit with all 240 exact integer compiled evaluations, seven complete finite structural checks, and Lean kernel-checked generic structural and minorant bridges. It includes the value and derivative tables, point lists, convexity chains, noncompact gap coverage and all 32 root trees. The evidence is recorded

in repository note 483 and its linked audits [7]. The compiled evaluations are not 240 kernel-reduced proofs, and the full upstream zeta formalization was not rebuilt. The present result has that explicit finite-computation trust scope.

The new majorant and energy checker uses only Python’s standard-library arbitrary-precision integers and rational arithmetic, with assertions enabled. It does not run the much larger native search supporting the numerical headline in [5]. In particular that search’s `native_decide` computation axiom is not an input to this paper. Neither this checker nor the earlier symbolic bridges constitute a complete Lean verification of Theorem 1.1. The mathematical transfer, finite computation and formalized sublemmas are separate evidence.

The new finite proof is recorded by `scripts/am_nine_point_epigraph_certificate.py` and `output/am-nine-point-epigraph-certificate.json`. The latter includes all 70 captured cells with their 26 atom records, 16 decoded Farkas integers, and all 289 sparse rational duals. The verifier reconstructs the original minorant guard before consuming the new cuts. Its `-replay` mode rebuilds the isolated capture from the fixed raw source and the two admitted generic bridges, checks all 41 old outer/root results, and compares every captured record. Its `-check` mode consumes saved records only. Both modes were actually run with assertions enabled; optimized Python is refused. The full earlier value and derivative admission is reused, rather than falsely described as repeated by the new 41 evaluations. The new certificate and analytic transfer were independently reviewed inside the project. Internal AI-assisted review is distinct from external peer review or complete kernel formalization.

Further improvement requires a stronger continuous local reward, a better window, or an actually paid additional spectral gain. The fixed-window method ceiling is not a bound on the true zero proportion. The present result does not improve the project’s separate zero-free-region bounds or pay its remaining original four-prime arithmetic covariance.

A Small exact rational certificate

The accompanying standard-library program is `scripts/am_lossless_majorant_certificate.py`, with report `output/am-lossless-majorant-certificate.json`. From the repository root run

```
python -B scripts/am_lossless_majorant_certificate.py -check.
```

The checker refuses optimized Python execution. The following specifies the full continuous bridge independently of floating-point output.

For a nonnegative rational interval $q \in [0, 10)$, it encloses $\cos \sqrt{q}$ and $\sin \sqrt{q}/\sqrt{q}$ by interval evaluation of

$$\sum_{k=0}^{12} \frac{(-1)^k q^k}{(2k)!}, \quad \sum_{k=0}^{12} \frac{(-1)^k q^k}{(2k+1)!},$$

with symmetric remainder bounds $q_{\max}^{13}/26!$ and $q_{\max}^{13}/27!$, respectively. The alternating tails decrease from the first omitted term. For the integer cosine modes the rational phase $2ju$ is reduced modulo two to $[-1, 1]$ before multiplication by π . No division by a small trigonometric argument is required. The normalization $Z_0 = \sin \beta/\beta$ has the exact argument squared $q = 1/2$.

Machin’s identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with the first 24 terms of each alternating arctangent series and its next-term remainder, certifies

$$3141592653/10^9 < \pi < 3141592654/10^9.$$

These enclosures, at all 129 nodes $u = k/256$, certify the strict lower margin $g(u) - f(u) > 9/100$. On all of I , $|\operatorname{sinc}'| \leq \pi/2$ follows from its integral representation. Thus

$$|(g - f)'| \leq 4\gamma\pi\theta + \frac{\sqrt{2} + 2\pi \sum_j j|c_j|}{Z_0} < 4\gamma\frac{22}{7}\theta + \frac{3/2 + (44/7) \sum_j j|c_j|}{9/10} < 16.$$

This proves the global margin $47/800$ stated above.

For (17), use the same Taylor intervals to bound $\cos \beta/Z_0$ from above, $Z_0^2 = 2 \sin^2 \beta$ from below, and π from above in the positive correction in (16). All resulting comparisons are exact rational inequalities. A shorter independent certificate uses

$$\begin{aligned} u &= \frac{3329448031}{4379443200} > \cos \beta, & z &= \frac{153778383027779}{167382319104000} < Z_0, \\ d &= \frac{180493}{213840} < 2 \sin^2 \beta, & p &= \frac{5277328977275528}{1679825970703125} > \pi. \end{aligned}$$

These arise, respectively, from cosine terms through index 6, sinc terms through index 7, cosine at $\sqrt{2}$ through index 6, and Machin arctangent upper/lower sums through indices 4 and 1. Direct rational arithmetic gives

$$\frac{3}{2} - \frac{u}{z} - \frac{\sum_j c_j^2(1/2 - 1/(4p^2 j^2))}{d} > \frac{67216841}{10^8}.$$

The positive margin is greater than $5.6 \cdot 10^{-10}$. The old checker also verifies the old span capacities, close-pair constants and majorant mass bound. The new rational ratio is checked by the accompanying nine-point verifier; the joint span budgets follow directly from (6) and the definition of a' . These small checks do not substitute for the full seven-dimensional certificate in Input 2.1.

References

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- [7] RH–Weil research project, public repository, note 483, the PC8 finite replay and independent continuous audits; notes 498–501 and the joint nine-point certificate.